Starting Over:
Learning, Fresh Starts and the Reallocation of Serial Entrepreneurs

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Motivation

- Many prominent examples of **serial entrepreneurs**, defined as someone who has run more than one business
  - 18–30% of entrepreneurs in Europe, about one in eight in the US (Plehn-Dujowich, 2010)
- Most firm dynamics models have no role for serial entrepreneurship
  - One productivity process per firm, and exit means leaving entrepreneurship (Hopenhayn, 1992; Hopenhayn & Rogerson, 1993)
  - Some exceptions (Plehn-Dujowich, 2010; Chen, 2013)
- Extensive focus on the role for supply-side policy, both academically and politically
- But a stylised fact is missing from the framework
  - In UK data, serial entrepreneurs run about one in five firms that can be linked to their owners, and those firms are more productive, so it is not just the Bond villains

Motivation

- Three questions:
  1. Why are serial entrepreneurs' firms more productive: **experience** or **selection**?
  2. Should starting over be included in existing firm dynamics models, and how?
  3. Does it change the role for policy?
     - Lowering the cost of starting over against raising the liquidation value of capital

Current State of Play

- Empirical emphasis on the age profile when considering firms' contribution to aggregate productivity growth and job creation
  - Young firms grow faster (Haltiwanger et al., 2017; Decker et al., 2017)
- Seminal firm dynamics models attribute any divergence in growth paths to ex-post heterogeneity (Hopenhayn, 1992; Hopenhayn & Rogerson, 1993)
- But ex-ante heterogeneity is present at the firm level (Campbell & De Nardi, 2009; Hurst & Pugsley, 2011; Queiró, 2022)
  - Ex-ante factors: over 90% of employment dispersion in the first year after entry, about 40% after twenty years (Pugsley, Sedláček & Sterk, 2021)
- Learning models have one productivity component, so a firm that learns it is poor can only exit (Jovanovic, 1982; Arkolakis et al., 2018)

Current State of Play

- Existing theories of entrepreneurship and firm dynamics treat exit as business failure and the end of entrepreneurship (Lucas, 1978; Jovanovic, 1982; Hopenhayn, 1992; Hopenhayn & Rogerson, 1993; Klette & Kortum, 2004; Luttmer, 2007)
  - Consequently the relevant margin for policy is entry and exit
- Empirically, extensive evidence against the idea that business formation is one and done (Lafontaine & Shaw, 2016; Shaw & Sørensen, 2019; De Vera et al., 2026)
- The cost of failing shapes which projects are started and abandoned (Gromb & Scharfstein, 2002; Landier, 2005; Manso, 2016)
  - More forgiving bankruptcy law goes with more business ownership (Fan & White, 2003; Armour & Cumming, 2008)

Existing Evidence

- **Denmark**, 2001–2013 (Shaw & Sørensen, 2019)
  - Serial entrepreneurs' firms have 98% higher sales than novices' and are 49% more productive
- **Texas**, new retail businesses, 1990–2011 (Lafontaine & Shaw, 2016)
  - 25.6% are operated by serial entrepreneurs
  - Prior business experience increases the longevity of the next business
- Similar magnitude of the serial share across other studies
  - UK 19–25%, US 12.5%, Germany 18% (Westhead et al., 2005; Headd, 2003; Wagner, 2003)

Existing Evidence

- **Portugal**, administrative data (De Vera et al., 2026)
  - Firms owned by serial entrepreneurs are larger, more productive, grow faster and are less likely to shut down

- Estimate:

$$y_{i,t} = \alpha + \beta \mathbf{1}_{SE} + \gamma F_{i,t} + e_{i,t}$$

  - Where $y_{i,t}$ = [firm size, probability of exit, firm growth, firm productivity]$'$, $\mathbf{1}_{SE}$ is a dummy equal to one if the firm is owned by a serial entrepreneur, and $F_{i,t}$ is a vector of controls

- The premia are already present in serial entrepreneurs' first businesses
  - Interpreted as selection on innate characteristics, rather than learning or luck

Recap

- Existence of serial entrepreneurs appears quantitatively important, in Portugal, Denmark, the US and the UK
- There has been renewed focus on the supply side over the business cycle
- Common feature of workhorse firm dynamics models: no ex-ante heterogeneity in the entrepreneur, and no option to start over
- The open question is whether the premium is created by experience or reflects who starts again

Roadmap

- Empirical evidence
- Model
- Counterfactuals
- Conclusion

Empirical Evidence

Data

- **Orbis Historical**, UK, fiscal years 2005–2024
  - Large, Medium and Small libraries pooled: revenue, employment, total assets, debt
  - 414,618 limited companies, 1.51 million firm–years
- **Companies House register of Persons of Significant Control**, from April 2016
  - Natural persons holding more than 25% of shares or votes
  - Person key: surname, first name, month and year of birth
  - 279,236 persons matched; 21,490 hold 2–20 panel companies $\Rightarrow$ serial entrepreneur
- Productivity:

$$\log \mathrm{TFP}_{i,t} = \log R_{i,t} - \mu^{-1}\left[(1-\alpha)\log n_{i,t} + \alpha \log k_{i,t}\right], \qquad \alpha = 0.30,\ \mu = \tfrac{\sigma}{\sigma-1} = \tfrac{4}{3}$$

Serial Entrepreneur Premia in the UK

	SE premium	s.e.
Log TFP	0.245	(0.008)
Log labour productivity	0.184	(0.009)
Employment growth (pp)	0.39	(0.11)
Exit hazard ratio	0.72	
Median size: regular / serial	4 / 38.9 employees	
Share of linkable firms	19.25%	

- Each outcome on a serial-entrepreneur dummy, with age dummies and 4-digit industry $\times$ year effects
- Compared with all other companies in the panel, including those not in the register
- Serial status does not vary within a firm, so these are comparisons across owners

Exit Declines with Age, and Most Firms Borrow Little

Age	Hazard $h(a)$	Survival $S(a)$
1	0.100	0.900
2	0.088	0.821
3	0.079	0.756
5	0.063	0.660
10	0.038	0.520
Pooled hazard, ages 1–10: 0.063		

- Serial-entrepreneur firms exit at 0.72 times the rate of regular firms of the same age
- Debt to total assets: mean 0.229, median 0.079; 32.5% of firm–years report no debt
- Each year 0.37% of companies enter insolvency, against 2.9% that close for any reason
  - Most exits are not defaults

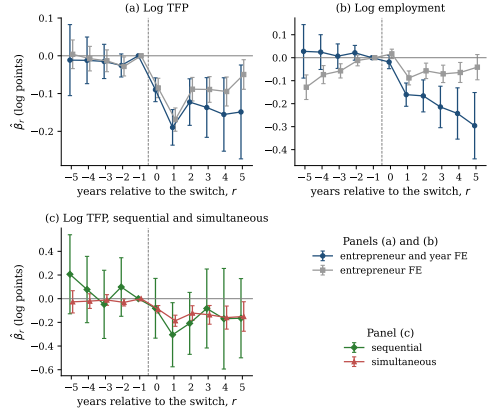
Over Half of Permanent Productivity Is the Entrepreneur's

- Permanent productivity = owner component + business component, nested random effects, method of moments corrected for sampling noise

Sample	Owner share	95% CI	Entrepreneurs
All owners	0.570	[0.553, 0.590]	11,236
Sequential	0.599	[0.483, 0.731]	307
Simultaneous	0.584	[0.564, 0.602]	10,372
Owners of two firms	0.587	[0.567, 0.604]	9,098
All owners, log TFP	0.566	[0.545, 0.585]	10,956

- Across an owner's two firms, average labour productivity correlates at 0.55

Productivity Does Not Rise at the Switch



- Switch = incorporation of the second business; $r = -1$ reference
- Flat before: coefficients between -0.028 and 0.004
- Falls while the new business starts up: -0.085, then -0.169
- $r \leq -3$ minus $r \geq 1$: 0.093 (0.016); 0.138 (0.069) with year FE

Navy: entrepreneur and year FE; grey: entrepreneur FE only

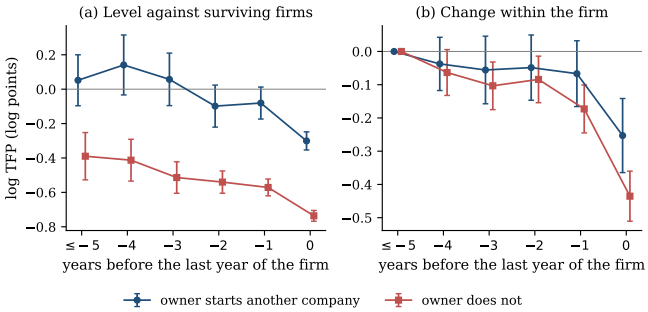
Later Businesses Are Not More Productive at the Same Age

	Log TFP	Log labour prod.
Second vs first business, ages 1–5	−0.085 (0.025)	−0.066 (0.026)
with industry FE	−0.028 (0.024)	−0.027 (0.025)
Order of the business, linear	−0.004 (0.015)	−0.005 (0.016)
Years of prior ownership	0.0027 (0.0010)	0.0024 (0.0011)
Second × years first business ran	−0.008 (0.010)	−0.010 (0.010)

- If running a business made an owner better, later businesses would be more productive at the same age

- Years of prior ownership raise log TFP by about 0.03 over ten years, which is small against a premium of 0.245

Owners Who Start Again Already Ran Better Firms



- Companies closed 2019–22: 24.4% of owners start or take over another company
- Last three years: log TFP 0.387 (0.030) higher than other closers, same age, industry, year
- Log employment 0.325 higher; owners 4.8 years younger
- Five or more years before: as productive as surviving firms (0.05, s.e. 0.08); others 0.39 lower

Summary of Empirical Results

- Serial entrepreneurs' firms have a log TFP premium of 0.245 and exit at 0.72 times the rate
- Exit declines with age; most firms borrow little, and only 0.37% a year enter insolvency
- The entrepreneur accounts for 56.6% of the permanent variance of log TFP
- Productivity does not rise when an owner starts a new business
- At the same age, later businesses are no more productive
- Owners who start again after a closure already ran more productive firms

$\Rightarrow$ The evidence points to selection on a permanent characteristic of the entrepreneur

Model

Model

Household

- Model the household very straightforwardly: a representative household that maximises

$$U = \sum_{t=0}^{\infty} \beta_H^t \ln Q_t, \qquad \beta_H = \frac{1}{1 + r_f}$$

- Where Q_t is a composite good aggregated from varieties $Q_{i,t}$:

$$Q_t = \left(\int Q_{i,t}^{\frac{\sigma-1}{\sigma}} \, di \right)^{\frac{\sigma}{\sigma-1}}$$

- Supplies $\bar{N}$ units of labour inelastically and owns all firms:

$$P_t Q_t = w_t \bar{N} + \Pi_t$$

- σ is the elasticity of substitution between varieties; the wage is the numeraire

Model

Firms

- Each variety i is produced by a firm operating business j with production function

$$y_{i,t} = Z_{i,t}\, k_{i,t}^{\alpha}\, n_{i,t}^{1-\alpha}$$

- Given CES demand, revenue is

$$R_{i,t} = \Lambda^{1/\sigma} X_{i,t} \left(k_{i,t}^{\alpha} n_{i,t}^{1-\alpha} \right)^{1/\mu}, \qquad \mu \equiv \frac{\sigma}{\sigma - 1}, \quad X_{i,t} \equiv Z_{i,t}^{1/\mu}$$

- $X_{i,t}$ is realised productivity in revenue units, which the data measure as TFP
- $\Lambda \equiv w^{1-\sigma} P^{\sigma} Y$ is the demand shifter

Model

Productivity

- The firm does not know its true productivity, but observes two signals each year:

$$\theta_{i,t} = \Theta_i + \varepsilon_{i,t}, \qquad \phi_{j,t} = \Phi_j + u_{j,t}$$

  - $\Theta_i = \Theta_i^K + \Theta_i^L$: **ability**, permanent, travels with the entrepreneur; $\Theta_i^K \sim N(\mu_\Theta, \sigma_K^2)$ known at entry, $\Theta_i^L \sim N(0, \sigma_L^2)$ learned
  - $\Phi_j \sim N(\mu_\Phi, \sigma_\Phi^2)$: **match quality**, redrawn when the entrepreneur switches
  - $\varepsilon_{i,t} \sim N(0, \sigma_\varepsilon^2)$, $u_{j,t} \sim N(0, \sigma_u^2)$ i.i.d.

- The two signals make up realised productivity:

$$X_{i,t} = \exp\left(\theta_{i,t} + \phi_{j,t} - \tfrac{1}{2}\left(\sigma_\varepsilon^2 + \sigma_u^2\right)\right)$$

Model

Learning

- Firms learn Θ^L and Φ from θ and ϕ (Jovanovic, 1982)
- Beliefs are normal and independent:

$$\tilde{\Theta}_{i,t} \sim N(A_{i,t}, B_x), \qquad \tilde{\Phi}_{j,t} \sim N(C_{j,t}, D_\tau)$$

- Where

$$A_{i,t} = \frac{x\sigma_L^2}{\sigma_\varepsilon^2 + x\sigma_L^2}\,\bar{\theta}_{i,x} + \frac{\sigma_\varepsilon^2}{\sigma_\varepsilon^2 + x\sigma_L^2}\,\Theta_i^K, \qquad B_x = \frac{\sigma_\varepsilon^2\sigma_L^2}{\sigma_\varepsilon^2 + x\sigma_L^2}, \qquad D_\tau = \frac{\sigma_u^2\sigma_\Phi^2}{\sigma_u^2 + \tau\sigma_\Phi^2}$$

  - x: ability signals observed; τ: tenure of the current business; $C_{j,t}$ analogous with prior μ_Φ

- Match variance halves after $\tau = \sigma_u^2/\sigma_\Phi^2$ signals, so the match is learned slowly when the noise is large
- At a switch: keep A and x; reset $C = \mu_\Phi$ and $\tau = 0$

Model

Labour and Profits

- Labour is chosen before output is realised (Arkolakis et al., 2018), on expected productivity

$$X(s) \equiv E\left[X_{i,t} \mid s\right] = \exp\left(A + C + \tfrac{1}{2}\left(B_x + D_\tau\right)\right), \qquad s = (A, C, x, \tau)$$

- Optimal labour choice:

$$n^*(s,k) = \left(\frac{1-\alpha}{\mu}\,\Lambda^{1/\sigma} X(s)\, k^{\alpha/\mu} w^{-1}\right)^{\frac{1}{1-(1-\alpha)/\mu}}$$

- Expected operating profit:

$$\pi^e(s,k) = \frac{\mu - 1 + \alpha}{\mu}\, R^e(s,k) - c$$

- Uncertainty raises expected productivity, so a new business hires more than an established one with the same means

Model

Timing

- Each period firms face the problem of whether to **continue**, **switch** to a new business, or **exit**

- Timing:
  1. The firm dies with probability δ_d; its capital is sold at the liquidation value ρ
  2. A surviving firm chooses, on its beliefs, to continue, switch or exit
  3. Continuing firms choose next-period capital and debt (k', b') with dividend $d \geq 0$, hire labour and produce
  4. Switching firms produce a last year, then close the business and repay the debt; the new business produces from next year
  5. Exiting firms do not produce, sell their capital at ρ, repay the debt and keep the surplus
  6. Signals θ and ϕ are realised and beliefs are updated

Model

Capital, Debt and Default

- One-period defaultable debt (Ottonello & Winberry, 2020); no equity issuance:

$$d = \pi^e(s,k) + (1-\delta_k)k - k' - \Psi(\iota,k) + q(s,k',b')\,b' - b \;\geq\; 0$$

- In default the capital is sold at ρ, the lender gets $\min\{b, \rho k\}$ and the owner keeps $V^D = \max\{0, \rho k - b\}$

- Risk-neutral lenders share the firm's beliefs and price the risk of exit or default next year:

$$q(s,k',b') = \frac{1}{1+r_f}\Big[(1-\delta_d)\Big(1 - E\big[p^D(s',k',b')\big]\,(1-\text{rec})\Big) + \delta_d\,\text{rec}\Big]$$

  - Lender's recovery in a default: $\text{rec} = \min\{1, \rho k'/b'\}$

- Debt up to $\rho k'$ is safe; beyond it the spread falls with the ability belief

Model

Switching versus Continuing

- A switching firm redeploys a fraction χ of its capital and sells the rest at ρ; the debt must be repaid:

$$e = \pi^e(s,k) + \rho(1-\delta_k)k - b \;\geq\; 0$$

- It keeps its ability belief and resets its match belief to the prior, $s^+ = (A, \mu_\Phi, x, 0)$:

$$V^S(s,k,b) = e + \chi(1-\rho)k - \lambda(s) + \max_{k',b'} \Big[-k' + q(s^+,k',b')\,b' + \beta\, E_{\theta|s}\big[\bar{V}(s^{+\prime},k',b')\big]\Big]$$

- $\lambda(s)$ is the **cost of starting over**, which covers time, legal, administrative and stigma costs and falls with the ability belief

$$\lambda(s) = \kappa\big[1 + \zeta\,(1 - A^{norm}(s))\big]$$

Model

Value Function

- The full value function, conditional on survival:

$$V(s,k,b) = S_\omega\Big\{ V^C(s,k,b),\ V^S(s,k,b),\ V^D(k,b) \Big\}$$

- Where

$$V^C(s,k,b) = \max_{(k',b'):\, d \geq 0} \; d + \beta\, E_{\theta,\phi|s}\big[\bar{V}(s',k',b')\big]$$

- S_ω is the smoothed maximum from extreme-value taste shocks:

$$S_\omega\{V_a\} = \omega \ln \sum_a e^{V_a/\omega}, \qquad \Pr(a \mid s,k,b) = \frac{e^{V_a/\omega}}{\sum_{a'} e^{V_{a'}/\omega}}$$

- The gain from a new business scales with $\exp(A)$ and $\lambda(s)$ falls with $A \Rightarrow$ switchers are selected on ability
  - Selection on match quality only to the extent firms learn about their match

Model

Entry

- There is a large mass of potential entrants, who pay an entry cost c_e
- After paying, an entrant learns the known part of its ability, Θ^K, so it enters at

$$s_E(\Theta^K) = \left(\Theta^K, \mu_\Phi, 0, 0\right)$$

  - Some knowledge of entrepreneurial ability, none of the match with this business
  - Buys capital $k_0(\Theta^K)$ at its frictionless optimum, with no debt

- The free entry condition pins down the demand shifter Λ:

$$V^E(\Lambda) = E_{\Theta^K}\left[(1-\delta_d)\, V\left(s_E, k_0, 0\right) + \delta_d\, \rho k_0 - k_0\right] = c_e$$

- Value is convex in the ability belief, so high-Θ^K entrants account for much of V^E and enter with more capital

Model

Equilibrium

- Stationary distribution built age by age from a unit inflow of entrants; entry equals exit
- Every outlay (capital, adjustment, c, λ, c_e) is produced one for one from labour; labour market clears:

$$\bar{N} = M(\bar{n} + o)$$

- Reallocation index, the CES mean of productivity implied by the posterior means of active firms:

$$X^{\sigma-1} \equiv E\left[\exp\left((\sigma-1)(A+C)\right)\right]$$

- Welfare: consumption-equivalent change between stationary equilibria, $Y_1 / Y_0 - 1$

Calibration

Targeted Moments

Moment	Data	Moment	Data
Serial-entrepreneur share	19.25%	Exit hazard, age 1	0.100
Log TFP premium	0.245	Pooled exit hazard, ages 1–10	0.063
Owner share of TFP variance	0.566	Five-year survival	66.0%
Event-study contrast	0.093	Mean leverage	0.229
Median employment, regular	4	Share of firm–years with no debt	32.5%
Median employment, serial	38.9	Insolvency rate	0.37%
SD of employment growth (pp)	28.43	Growth premium, age-controlled (pp)	0.39
		Growth difference, unconditional (pp)	0.22

- At the estimate: $\rho = 0.40$, $\zeta = 12.1$, $c_e = 269.5$; the model gives a serial share of 0.189 and a premium of 0.248

Counterfactuals

Counterfactual Experiments

- Three sets of experiments at the estimate:
  1. Lower the cost of starting over: cut κ by a quarter, by a half, or to zero, holding ζ fixed
  2. Change the liquidation value of capital: ρ from 0.40 to 0.50, or down to 0.30
  3. Give firms full information about ability, match, or both

- Entry cost held at $c_e = 269.5$; solve for Λ^* with $V^E(\Lambda^*) = c_e$. Output moves with labour in production and the demand shifter:

$$\Delta \ln Y = \mu \, \Delta \ln N - \frac{\Delta \ln \Lambda^*}{\sigma - 1}$$

- Welfare is the consumption-equivalent change, which equals the change in output, with $\lambda(s)$ either a **resource cost** (uses labour) or a **private cost** (stigma, no labour)

The Cost of Starting Over and the Liquidation Value

		Base cost κ			Liquidation value ρ	
	Baseline	$\times 0.75$	$\times 0.5$	0	0.50	0.30
Demand shifter Λ^*	1.000	0.995	0.988	0.916	0.994	1.005
Output Y		+0.11	+0.13	+2.01	+0.26	−0.22
Welfare, resource cost (%)		+0.11	+0.13	+2.03	+0.26	−0.22
Welfare, private cost (%)		+0.18	+0.41	+1.76	+0.28	−0.23
Reallocation index $\mathcal{X}$		−0.20	−0.64	−3.21	+0.10	−0.16
Mass of firms (baseline $= 1$)	1	1.007	1.039	1.209	0.996	1.002
Switching per 100 firms	1.25	1.95	4.11	33.67	1.34	1.18
Serial-entrepreneur share	0.189	0.272	0.458	0.939	0.199	0.181
Log TFP premium	0.248	0.147	−0.014	−0.054	0.246	0.243
Mean employment per firm	105.8	105.0	101.6	86.9	106.2	105.5

- Output, index and capital: 100 $\times$ log difference from the baseline; free-entry general equilibrium

Lowering the Cost of Starting Over Adds Firms

- Cutting κ by a quarter: switching from 1.25 to 1.95 per 100 firms, serial share from 0.189 to 0.272, welfare up by 0.11% to 0.18%
- The premium falls from 0.248 to 0.147
  - With $\zeta = 12.1$ a proportional cut removes the most cost for the least able owners, so the additional switchers are less able
  - Switchers' ability belief exceeds the average by 0.075, against 0.163 at the baseline
- Removing κ: 34 switches per 100 firms, serial entrepreneurs run 94% of firms
  - Value of entry rises, Λ^* falls to 0.916, firms are smaller (86.9 against 105.8 employees), and 21% more of them
  - Output and welfare rise by 2.0% and the reallocation index falls by 3.2%, so the gain comes from the number of firms

Raising the Liquidation Value Works Through Productivity

- ρ sets the value of capital that a switcher does not redeploy, of the whole capital stock at exit, default or death, and of the collateral lenders price
- Raising ρ from 0.40 to 0.50: output and welfare rise by 0.26% with a resource cost and 0.28% with a private cost
  - More than cutting κ by a quarter, under either treatment
- Switching rises only to 1.34, and the premium is almost unchanged, at 0.246 against 0.248
  - The cost falls in proportion to capital, $(1-\rho)(1-\delta_k-\chi)k$, and the able run the largest businesses, so the additional switchers are as able as the others
  - Switchers' ability belief exceeds the average by 0.169, against 0.163
- Higher ρ also raises the value of exit and default, and insolvent exits rise from 0.148 to 0.162 per 100 firms

Learning Against Full Information

		General equilibrium		
	Learning	Full info	Ability known	Match known
Median age at first switch, top quartile	8	3	14	3
Share of top quartile switched by age 3	0.167	0.379	0.077	0.382
Ability gap, switched minus not, age 3	0.206	0.534	0.481	0.428
Switching per 100 firms	1.25	0.90	0.68	1.89
Event-study contrast	−0.085	−0.043	−0.139	−0.020
Demand shifter Λ	1	0.808	0.889	0.907
Labour in production (baseline $= 1$)	1	0.948	0.947	1.002
Reallocation index $\mathcal{X}$		+5.65	+0.74	+4.69
Output Y		−0.07	−3.42	+3.57
Welfare, resource cost (%)		−0.07	−3.36	+3.64
Welfare, private cost (%)		−0.14	−3.46	+3.71

- No policy delivers this information, so the regimes measure the role of learning in the model

Learning Delays the Switches of the Most Able

- With full information the most able quarter first switch at a median age of 3 against 8 under learning, and switching is more closely tied to ability (gap 0.53 against 0.21)
  - Fewer of them ever switch, 0.64 against 0.72, suggesting that under learning able owners also leave good matches, since they cannot tell them from poor ones

- However, full information leaves welfare almost unchanged in general equilibrium, at -0.07%
  - Entrants who know their ability enter larger, so more labour produces capital goods and labour in production falls by 5.2%
  - Value of entry rises, Λ^* falls 19%, offsetting almost all of the loss

- Knowing only the match raises welfare by 3.6%, while knowing only ability lowers it by 3.4%
  - A known match lets owners in poor matches switch early, worth most to the able, who run the largest businesses
  - Profit at the frictionless scale rises with X^4, so a match one s.d. better (0.042 in logs) raises it by about 18%

Decomposition of Output

- $\Delta \ln Y = \underbrace{\Delta \ln \mathcal{X}_B + \Delta \ln \mathrm{AE} + \Delta \ln \mathrm{IM}}_{\text{productivity}} + \alpha \Delta \ln K + \frac{\Delta \ln M}{\sigma - 1} + (1 - \alpha) \Delta \ln N$

	Base cost κ			Liquidation value ρ	
	$\times 0.75$	$\times 0.5$	0	0.50	0.30
Productivity	$+0.01$	-0.48	-2.98	$+0.30$	-0.15
efficient index $\mathcal{X}_B$	-0.08	-0.48	-3.16	$+0.28$	-0.09
allocative efficiency	$+0.05$	-0.02	$+0.07$	$+0.01$	-0.03
input mix	$+0.04$	$+0.02$	$+0.12$	$+0.01$	-0.02
Capital deepening $\alpha \Delta \ln K$	-0.09	-0.52	-0.85	$+0.05$	-0.11
Variety $\Delta \ln M/(\sigma - 1)$	$+0.23$	$+1.26$	$+6.32$	-0.12	$+0.06$
Labour in production $(1 - \alpha) \Delta \ln N$	-0.04	-0.14	-0.49	$+0.04$	-0.02
Total	$+0.11$	$+0.13$	$+2.01$	$+0.26$	-0.22

- Cutting κ raises output through the number of firms, and raising ρ through productivity

Conclusion

- The serial-entrepreneur premium points to selection on a permanent characteristic of the entrepreneur, and running a business adds about 0.03 to log TFP over ten years
- In the model, switchers are selected on ability, and learning delays the switches of the most able, whose median age at first switch is 8 against 3 with full information
- **Policy:**
  - Cutting the cost of starting over adds firms but brings in less able switchers, raising output by 0.11% for a quarter cut and 2.0% if removed
  - Raising the liquidation value from 0.40 to 0.50 raises productivity and output by 0.26%, more than a quarter cut in κ
- Not observing the match costs more than the cost of starting over (3.6% against 2.0%); not observing ability at entry raises welfare once entry adjusts